25900405

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Core Course (DSC) - MG3DSCFSC201 – FORENSIC CHEMISTRY
(2024 ADMISSION ONWARDS)
Scheme for evaluation

PART A – Short Answer Type Questions (2 marks) each

1. The chain of custody is the documented and unbroken transfer record of evidence from the scene to the court, ensuring its integrity and admissibility. (2 marks)
2. **Chromotropic acid test** detects formaldehyde (formed by oxidation of methanol). In presence of **conc. H_2SO_4**, chromotropic acid reacts with formaldehyde to give a **purple/violet colour**. (2 marks)
3. Determining the **point of origin** helps identify whether the fire was **accidental or deliberate**, and assists in locating accelerants or ignition sources. (2 marks)
4. **Rf (Retention factor)** represents the ratio of distance travelled by the compound to that by the solvent front; indicates **component identity and purity**. (2 marks)
5. 1. Wear lab coat, gloves, and goggles. 2. Avoid direct inhalation of fumes or mouth pipetting. Any two relevant safety measures- (2 Marks)
6. A **non-alcoholic beverage** is a drink containing less than 0.5% alcohol by volume, e.g., fruit juice, soft drinks. (2 marks)
7. Fats retain **absorption patterns of substances** and can indicate contact with accelerants or human origin (through lipid profiling). (2 marks)
8. Different inks show **distinct absorption spectra** under UV light, helping in **identifying alterations or different pen types**. (2 marks)
9. TLC is based on **differential adsorption** of compounds on a stationary phase (silica/alumina) and their movement with a mobile phase (solvent). (2 marks)
10. **Congeners** are minor components like esters, aldehydes, and higher alcohols that affect the **flavour, aroma, and toxicity** of beverages. (2 marks)
11. Because **physical properties** (viscosity, colour) and **chemical composition** (additives, hydrocarbons) together identify the source or brand of oil. (2 marks)
12. In **trap cases**, phenolphthalein powder is applied to currency; on contact with **NaOH**, it turns **pink**, confirming bribe acceptance. (2 marks)

Part B

Short Essay Type Questions

Answer any 5 questions

13. **Rhodamine test** detects synthetic dyes. The dye extracts give **fluorescent pink coloration** under UV when treated with rhodamine B reagent — used to compare textile or fibre dyes. (Principle – 1, Procedure – 2, Forensic use – 1)

14. In **SPE**, the analyte is retained on a solid adsorbent column; impurities are washed off, and analyte is eluted. Used for **drug purification** before GC/MS or HPLC analysis. (Mark distribution- Principle – 1, Procedure/steps – 2, Forensic use – 1)

15. Fractions include **gasoline, kerosene, diesel, lubricants, asphalt**—each used in transport, energy, and industrial products (mark distribution-Identification – 2, Use – 2)
16. Oils indicate **contact or contamination**; fats (solid) may indicate **biological origin**. TLC and GC analysis confirm source (animal, vegetable, mineral). (Differences – 2, Analysis – 1, Value – 1)
16. Definition of IMFL and Non-alcoholic beverages. IMFL: Alcohol > 40%, distilled spirits (whisky, rum). Non-alcoholic: <0.5%, no intoxication. (Difference – 2, Example – 2)

17. Definition – 1 marks Eg-Gasoline – motor fuel; Kerosene – lamps; Diesel – engines; Lubricants – machinery; Asphalt – roads. (three examples 3 marks)

18. Non relevant question – Any genuine attempt which is valid may provide with full marks

Part C
Essay Type Questions
Each question carries 10 marks

19. - **Cations:** NH_4^+, K^+, Na^+ indicate ammonium nitrate, potassium chlorate, etc. (2.5 marks) - **Anions:** NO_3^-, ClO_3^-, SO_4^{2-} indicate oxidizers (2.5 marks).
    - Identified by tests like **luminol**, **Griess test**, and **ion chromatography**.
    - Helps establish **type of explosive** and **source**. Analysis methods-any relevant (5 marks)

20. - Country liquor: legally manufactured traditional spirits. - Illicit liquor: illegally distilled, may contain methanol. - Denatured spirit: ethanol with toxic additives to prevent drinking. - IMFL: branded distilled beverages. - Non-alcoholic: <0.5% alcohol. - Discuss chemical composition, legal status, and health impact. (Definition – 3, Differences – 5, Legal aspect – 2)